

ENEL301 - Fundamentals of Engineering
Economics and Management

Introduction to Engineering Practice & Ethics

The background....

1. Engineers cannot achieve much without a lot of help from other people.
Collaboration drives performance and value.
2. Nothing is possible in engineering without money, usually lots of it.
3. A lot of engineering endeavours fail or end-up with major problems; 1 in 6 projects end-up with investors losing all of their money.
 - Poor collaboration is the #1 reason for project failure (or in other words, collaboration is the #1 reason for success)
 - Culture and communication are key determinants for successful collaboration

The background....

4. We need to do the right thing
 - Social culture shapes us, and therefore our ethics → everyone is different
 - You need to follow the law (but what does that mean?)
 - Engineers must ensure sustainable resource use
5. Engineers need to make things happen, all while managing uncertainty.
 - Uncertainty creates risk
 - Risk management is how we manage the uncertain
 - We need to account for people's views in the work that we do → value and acceptance

What industry has told us what you need to be able to do....

1. Apply ethical principles within engineering contexts, and recognise the social and cultural factors which affect ethics.
2. Apply individual management and inclusive teamwork practices in New-Zealand and global engineering contexts.
3. Conduct financial analysis in engineering contexts, including estimation of cost, cash-flow, financial balances, and financial viability.
4. Evaluate the viability of engineering projects, considering social, ethical, national and international laws, environmental, and cultural factors.
5. Implement engineering project and risk management techniques, including identification of- and communication with-, diverse stakeholders.

Timetable

Lectures

Tuesday 1 pm – 2 pm A1

Friday 10 am – 11 am A1

Tutorials / Workshops

Tutorial A. Week 1, and Weeks 4 to 12:

Wednesday 3 pm – 4 pm OR

Wednesday 4 pm – 5 pm

Bicultural Competency and Confidence Workshops

Tutorial B: Weeks 2 or 3:

Wednesday 3 pm – 5 pm Check your timetable for room

**Make sure you allocate yourselves
by the end of this week.**

**If you have a clash – let me know by
email**

**I will be forming teams based on
your Tutorial A room**

Lecture recording

- Lectures will be recorded on Echo360 & will be available on LEARN
- Tutorials / workshops will not be recorded → substantial discussion / interaction
 - Turn up to get the most out of these, some of what we will discuss will be on tests and exams

Assessment

Item	% of final grade	Individual / group	Due date
Bicultural confidence and competence workshop. Attendance	1%	Individual	Wednesday 22 July 15:00 – 17:00 (B01 – 07)
			Wednesday 29 July 15:00 – 17:00 (B08 – 14)
Pepeha	4%	Individual	Friday 31 July 17:00 (B01 – 07)
			Friday 07 August 17:00 (B08 – 14)
Test	25%	Individual	Tuesday 08 September 18:30 – 19:30 <i>C1 & C2 Lecture theatres. Refer to your timetable</i>
Sustainability assignment	20%	Group	Friday 09 October 17:00
Exam	50%	Individual	Examination period

Assessment

- BiCC workshop attendance is compulsory. If you don't attend, you won't receive the 1% grade
- Pepeha and Sustainability Assignment - LEARN submission (subject to TurnItIn)
- Late penalties of 10% (in absolute terms) per day or part day late will be deducted from the original mark. For example, an assignment with a nominal mark of 83% submitted 0-24 hours late will receive a mark of 73%, and submitted 24-48 hours late will receive 63%
- Request for extensions must be made to course coordinator
- Extensions for group assessment not normally given
- Special Consideration if you cannot sit the test and or/exam; 5 day limit to apply
- You will be permitted to use GenAI to assist you for some assessment items; check each item for more details
- We will endeavour to grade your assessment and provide you feedback within two weeks

Assessment

Hurdle requirement for Test and Exam

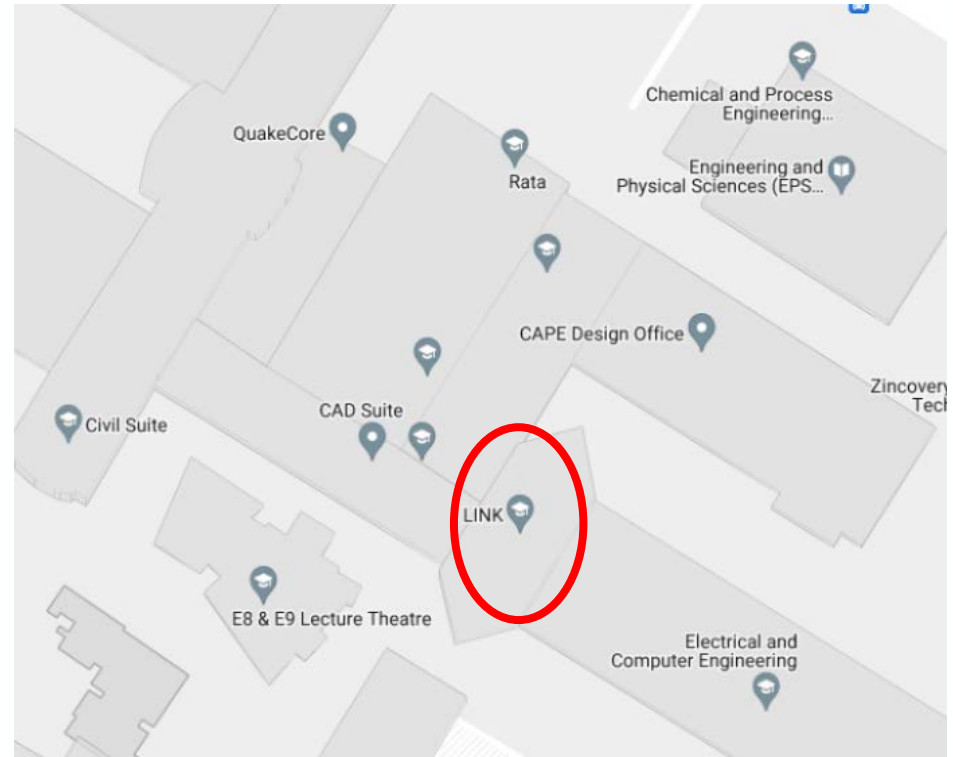
- The examiners will award a failing grade to students who score less than 40% for the Tests and Exam combined. More formally, $(\text{TestPercent} * 0.25 + \text{ExamPercent} * 0.5) / 0.75$ must be greater than or equal to 40 for a pass mark to be awarded.

Your teaching team

Course coordinator and lecturer



- Link A506
- Appointments Monday and Tuesday 9 – 9:45; book through LEARN
- Post content and assessment questions on the forum
- Need a different time? Email me: enda.crossin@canterbury.ac.nz



Your teaching team

Guest lecturers

Associate Professor Matthew Barber
Faculty of Law
Lecturer for “Law in New Zealand”



Virginia Nicholls
Senior Associate, Corcoran French
Lecturer for “Intellectual Property”

Contact Enda Crossin if you
have questions on this content

Your teaching team - Week 1 & 4-12 workshops

Weeks 1, 4-12	Session	Room	Tutors
Wed 15:00 - 16:00	A-01	Rātā 222 and 223	Cam Mantell
			Chris Huynh
			Felix Blanchard
			Gabriella Eastgate
			Hannah Botting
	A-02	Rāta E12	Gabby Rosemergy
			Kahu Jones
Wed 16:00 - 17:00	A-04	Rātā 222 and 223	Cam Mantell
			Chris Huynh
			Felix Blanchard
			Gabriella Eastgate
			Hannah Botting
	A-03	Rāta E12	Gabby Rosemergy
			Kahu Jones

Your teaching team

Bicultural Confidence and Competence

Week	Session	Room	Tutor
Week 2 Wed 22 July 15:00 - 17:00	B-01	E12	Stella Keenan
	B-02	Jack Erskine 340	Hēmi McNeill
	B-03	Jack Erskine 101	Sam Draper
	B-04	Jack Erskine 111	Lydia Frater
	B-05	John Britten 117 HP	Gabby Rosemergy
	B-06	Meremere 526	Kahu Jones
	B-07	Elsie Locke 104A	Benjamin Cosford
Week 3 Wed 29 July 15:00 - 17:00	B-08	E12	Stella Keenan
	B-09	Jack Erskine 340	Hēmi McNeill
	B-10	Jack Erskine 101	Sam Draper
	B-11	Jack Erskine 111	Lydia Frater
	B-12	John Britten 117 HP	Gabby Rosemergy
	B-13	Meremere 526	Kahu Jones
	B-14	Elsie Locke 104A	Benjamin Cosford

Feedback from last year...

The good...

- Teaching team were helpful, approachable, and responsive
- Lectures and workshops were sometimes useful and linked together well
- Overall workload felt about right
- Appreciated the efforts to improve the course

The bad

- Sustainability assignment confusing, not enough guidance in tutorials
- LEARN difficult to navigate
- Heavy workload at the end (Assignment)
- Test was too easy

What's changing?

- Make the assignment clearer and better supported
- Reorganised LEARN & reduced jargon for improved navigation
- Releasing Sustainability Assignment earlier & better integration with tutorials
- Increasing difficulty in Test

What we are going to cover and when...

Week	Lectures Tuesday 10-11 & Thursday 9-10	Tutorial / Workshop Weeks 1, 4-6 Wednesday 3-4 or 4-5. Weeks 2&3 Wednesday 3-5
1	Introduction to Engineering Practice & Ethics	Ethics
	Collaboration & Teamwork	
2	Māori Perspectives in Engineering	Bicultural Competence and Confidence (timetable and attend one of B01-07 OR B08-14)
	Bias, Diversity & Ethics	
3	Intellectual Property (1/2)	
	Intellectual Property (2/2)	
4	Law in New Zealand (1/2)	IP Protection
	Law in New Zealand (2/2)	
5	Data Privacy and Data Ethics	Law, Data Privacy, Data Ethics
	Business Structures, Strategy, Decisions & Risk	
6	Sustainable Development & Systems Thinking	Sustainable Development & Systems Thinking
	Life Cycle Assessment (1/2)	

What we are going to cover and when...

Week	Lectures Tuesday 1-2 & Friday 10-11	Tutorial / Workshop Weeks 7-12 Wednesday 3-4 or 4-5
7	Life Cycle Assessment (2/2)	Life Cycle Assessment
	Social and Economic Assessment	
8	Time Value of Money & Cash Flow	Time Value of Money & Cash Flow
	Cost Accounting	
9	Financial Accounting & Reporting (1/2)	Cost Accounting, Financial Accounting, Financial Reporting
	Financial Accounting & Reporting (2/2)	
10	Project Management	Project Management
	Risk Management	
11	Opportunity Analysis	Support for Sustainability Assignment
	Integrated Decisions (1/2) Guest Lecturer – Mike Ullrich (Fisher & Pykel Technologies)	
12	Integrated Decisions (2/2)	Risk Management and Opportunity Analysis
	Course Summary	

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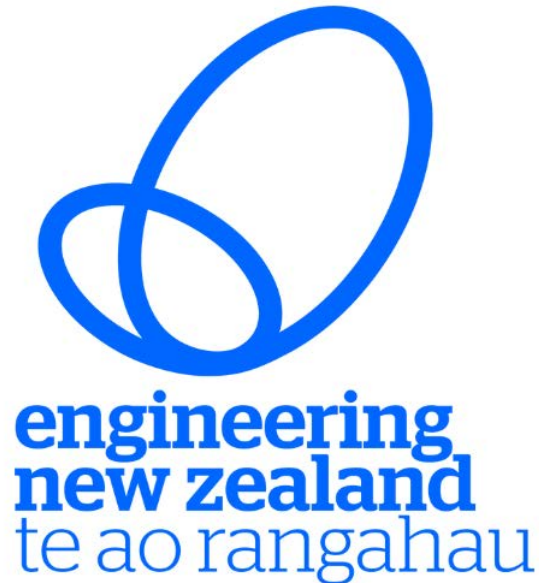
Ethics

(revision from ENGR101)

What industry has told us what you need to be able to do....

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3. Conduct financial analysis in engineering contexts, including estimation of cost, cash-flow, financial balances, and financial viability.
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5. Implement engineering project and risk management techniques, including identification of- and communication with-, diverse stakeholders.

We need engineers who “do the right thing”



“Engineers’ work affects all New Zealanders, every day. Engineers create the foundation for New Zealand’s social, environmental and economic fabric. ”

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At least 3 killed

Author: Ajzh2074

Boeing 737 MAX (Indonesia 2018 & Egypt 2019)

Source: PKREN, https://commons.wikimedia.org/wiki/File:Lion_Air_Boeing_737-MAX8_PK-LQP.jpg



Ethics: The Study of Moral Principles

- From êthos (ἦθος) [2] 'character' and 'personal disposition'
- What ought (should) to be done
- What character a person ought to be
- What values and morals a person ought to adopt



Ethics and the Law

- Ethics: what a person **ought** to do
- Law: what a person is **required** to do by the state
- Law establishes **minimum standards of conduct**
- Can be expanded to expectations required of institutions, e.g. Engineering New Zealand

Applied ethics

Goal: to make ethically sound decisions

1. Personal sense of right and wrong
 2. **Understanding of ethical theories**
 3. Systematic decision-making process
- **Rule-Based Thinking**
 - Do the “right” thing
 - Deontology
 - **Ends-Based Thinking**
 - Get the “best” outcome
 - Consequentialism
 - **Virtues-Based Thinking**
 - Be a “good” person
 - Virtue Ethics

Rule-Based Thinking

Deontology

- What are the “Rules”*?
- Follow the law
- Treat others as you would wish to be treated (the Golden Rule)
- Follow your professional code of ethics
- Obey company/organisational policies
- People have fundamental **rights** that other people have a **duty** to respect
- Actions must be **just** and **fair**

* Subject to change!

Rule-Based Thinking

The Golden Rule

- Treat others as you would wish to be treated
- "None of you [truly] believes until he wishes for his brother what he wishes for himself." - 40 Hadith Nawawi 13
- "Do to others as you would have them do to you" - Luke 6:31, Matthew 7:12
- "This is the sum of the Dharma [moral law / responsibilities / duties]: do naught unto others which would cause you pain if done to you." - Mahabharata 5:1517

Shortcomings

- Assumes your needs and wants are the same as everyone else's
- Does not distinguish between competing stakeholders

Ends-based thinking

Consequentialism

- Weigh competing actions according to their consequences
- Acts are ethical when they achieve the best outcome
 - “The ends justifies the means”
- Utilitarianism
 - Acts are ethical when they achieve the greatest good for the greatest number
 - Cheaper to pay for damages rather than repair....



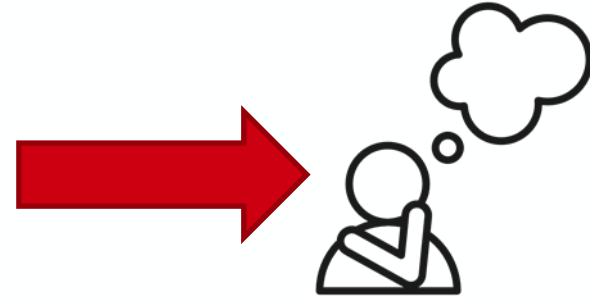
Virtues-Based Thinking

- Acts are ethical if they are what a “virtuous” person would do in the same circumstances.
- A virtuous person would act with
 - Honesty
 - Courage
 - Compassion
 - Self-regard
 - Generosity
 - Tolerance
 - Love
 - Fidelity
 - Integrity
 - Fairness
 - Self-control
 - Prudence

Applied ethics

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3. **Systematic decision-making process**



Applied Ethics - Blending ethical theories

Why & How?

- People do not have to align themselves with a single theory
 - Do not have to choose between being a strict deontologist or a strict teleologist
 - e.g. rule consequentialism: follow the rules, but when they conflict, choose based on the consequences

Applied Ethics - Blending ethical theories

Why & How?



“The Code of Ethical Conduct sets out the shared expectations of ethical practice that engineers have agreed to uphold. It’s a benchmark that confirms the expected standards of conduct for members....

.... The Code of Ethical Conduct is designed to help you make an ethical decision, and the decision that is best aligned to the standards of ethical behaviour set by your profession.”

Engineering NZ

A few closing words and some reading if you are interested

- Engineering designs are a result of a series of choices; choices are affected by your values
- Engineering designs are never ethically 'neutral'
- Challenging the way engineering ethics should be taught.... Lawlor, R. (2021). Teaching engineering ethics: A dissenting voice. *Australasian Journal of Engineering Education*, 26(1), 38-46.
<https://doi.org/10.1080/22054952.2021.1925404>

Preparation for the rest of this week

Before this week's workshop:

- Under the **Introduction to Engineering Practice & Ethics** Section
- Watch the ethical theory recaps from ENGR101 on LEARN
- Review the workshop material. Read the IEEE Code of Conduct

Before next lecture:

- Under the **Collaboration & Teamwork** Section
- Listen to this podcast (takes about an hour): [S5E22 James Trevelyan | What Research Tells Us Is The Most Important Quality of Engineers](#)
- Spotify and Apple hyperlinks also available via LEARN

Be ready for some conversations!

What's next?

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5	Data Privacy and Data Ethics	Law, Data Privacy, Data Ethics
	Strategy, Decisions & Risk	
6	Sustainable Development & Systems Thinking	Sustainable Development & Systems Thinking
	Life cycle assessment (1/2)	